

Potemkin PaaS

June 2010

Version 1.0

Potemkin because...

No real business problem was being solved.



The purpose of the exercise was to try things out (and show things off).

The Objective(s) and the outcome



A scale out run time fabric – run stuff, and not care where:

- Private

- Public

- Hybrid

also to learn about:

- Virtual Appliance factories

- Overlay networks

- Automated deployment (and scale up/down)

First – take a scalable run time fabric Capital SCF

The Fabric:

- Java – a language I'm comfortable working with.
- OSGi – a means to describe fine grained dependencies.

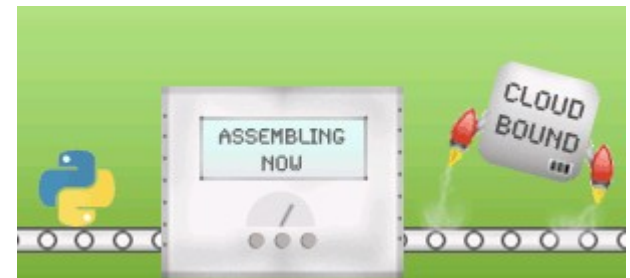


What I did with it:

- Made a virtual appliance that assembled an application server and a demo application from a bundle URL

Then

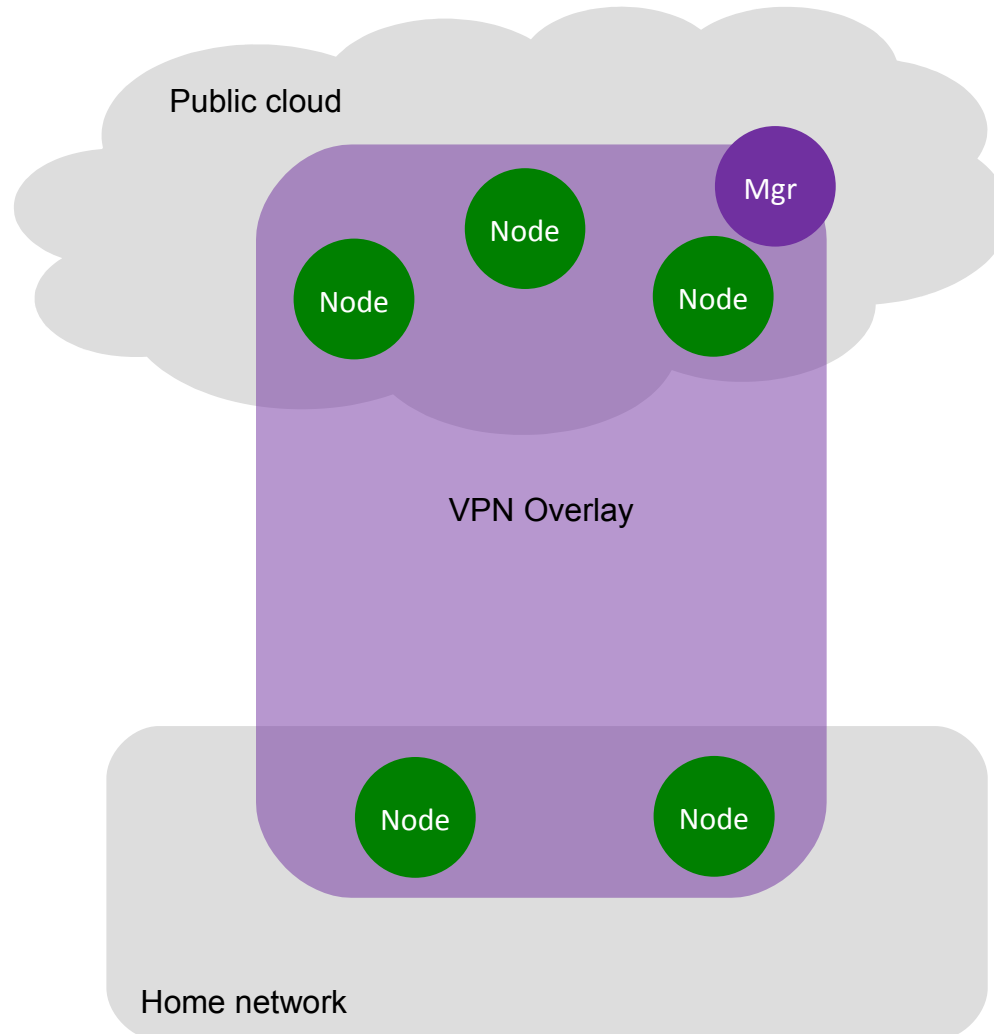
- Made a virtual appliance that could be a node in a fabric



But...

- Multicast needed so that nodes can discover each other and become a fabric

add – a multicast enabled overlay



but this gets tedious - fast

Start up VPN overlay manager

Download client configuration packs

Start up node 1

Install client configuration pack for node 1

Join node 1 to the overlay

Start up node 2

Install client configuration pack for node 2

Join node 2 to the overlay

Start up fabric on nodes 1 & 2

Let nodes 1 & 2 discover each other

.....

so automate, automate, automate



‘Boot scripts as a service’

- As each node comes up it calls in to the ‘context’ service
 - Who am I?
 - What should I do?
 - How do I connect to others like me?

Allowed the launch of the entire fabric to be boiled down to a single command line.

A terminal window titled 'root@jaunty: /opt/configcubed' with a red arrow pointing to the command prompt. The command entered is './cluster start ./clusters/atlas.xml'. The output shows the AWS account and endpoint, followed by a separator line, then 'Launching group: Atlas...', another separator line, and finally the details of the launch: 'Launching [redacted]', 'Instance Type: m1.small', 'UserData: servername=atlasserver role=atlas_endpoint clustername=atlas1 cont extmanager=[redacted].compute-1.amazonaws.com', 'Instances: 1 min, 1 max', and 'Security group: "vpncubed-client"'.

```
root@jaunty: /opt/configcubed
root@jaunty:/opt/configcubed# ./cluster start ./clusters/atlas.xml
Using AWS Account: [redacted]
Using Endpoint: https://us-east-1.ec2.amazonaws.com

=====
Launching group: Atlas...
=====

Launching [redacted]
Instance Type: m1.small
UserData: servername=atlasserver role=atlas_endpoint clustername=atlas1 cont
extmanager=[redacted].compute-1.amazonaws.com
Instances: 1 min, 1 max
Security group: "vpncubed-client"
```

Thanks for your attention



Gory details are at <http://wp.me/palij-3O>

Thanks also to:



Pat Kernan



Richard Nicholson



